Kazi Tahsin Mahmood

Assistant Teaching Professor
School of Engineering, Bowling Green State University

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SUMMARY

Ph.D. in mechanical engineering specializing in classical mechanical and acoustic systems that emulate quantum-inspired computations, topological computing, and nonlinear wave dynamics.

- Develop theoretical, numerical, and experimental frameworks for studying state formation, phase evolution, modal superposition, and topological behavior in coupled elastic and acoustic systems.
- Design and validate elastic-bit and phi-bit platforms that use nonlinear vibration, harmonic response, and phase-controlled dynamics to realize classical analogues of quantum states, gates, and multi-bit interactions.
- Research experience on topological acoustic and quantum-inspired mechanical networks, with applications in robust sensing, analog information processing, acoustic waveguides, and scalable physical-state platforms.

ACADEMIC CREDENTIALS

Wayne State University **Detroit, Michigan, USA**
Ph.D. *in Mechanical Engineering* *Jan 2022 – Dec 2025*
Dissertation Title: *Classical Realization of Quantum and Topological States through Nonlinear Elastic Networks*
Committee: Dr. M. Arif Hasan (Chair); Dr. Sean Wu; Dr. Chin-An Tan; Dr. Mohammad Bukhari; Dr. Pierre Deymier

Wayne State University **Detroit, Michigan, USA**
M.Sc. *in Mechanical Engineering* *Jan 2022 – May 2025*
Thesis Title: *Nonlinear-Dynamics-Driven Quantum Analogues in Classical Granular Metamaterials*
Advisor: Dr. M. Arif Hasan

Bangladesh University of Engineering & Technology **Dhaka, Bangladesh**
B.Sc. *(Engg.) in Mechanical Engineering* *Feb 2016 – Mar 2021*
Thesis Title: *Electro-Thermo-Mechanical Response of a Thin Conductive Layer of Metal Matrix Composite Reinforced by Graphite Fibers*
Advisor: Dr. Shaikh Reaz Ahmed

RESEARCH INTERESTS

Topological Metamaterials	Quantum-Analogous Computing	Nonlinear Dynamics
Acoustic Waveguides	Vibrational Mechanics	Solid Mechanics

PUBLICATIONS

1. **Kazi T. Mahmood** and M. Arif Hasan, "Geometric-phase diagnosis of mass defects through topological states in elastic granules," *Chaos, Solitons & Fractals*, vol. 212, pt. 2, p. 119024, 2026.
2. **Kazi T. Mahmood** and M. Arif Hasan, "Harnessing Nonlinear Dynamics for Time-Driven Berry Phase in Classical Systems," *Nonlinear Dynamics*, vol. 114, p. 689, May 2026.
3. **Kazi T. Mahmood**, M. Afridi Hasan, Abrar N. E. Faiaz, M. Arif Hasan, Pierre A. Deymier, Keith Runge, and Joshua A. Levine, "Multi-Bit Quantum-Inspired Dynamics in Nonlinear Mechanical Oscillators," *Journal of Applied Mechanics*, vol. 93, no. 6, p. 061003, May 2026.
4. **Kazi T. Mahmood** and M. Arif Hasan, "Topological Vibration Analysis of Elastic Lattices via Bloch-Sphere Mapping," *Journal of Vibration and Acoustics*, vol. 148, no. 4, p. 041004, Feb. 2026.
5. **Kazi T. Mahmood**, Abrar N. E. Faiaz, M. Arif Hasan, Pierre A. Deymier, Keith Runge, and Joshua A. Levine, "Experimental realization of logical elastic bits as qubit analogues in a nonlinear oscillator," *Scientific Reports*, vol. 16, no. 1, p. 3398, Dec. 2025.

6. **Kazi T. Mahmood** and M. Arif Hasan, "Topological insights from state manipulation in a classical elastic system," *AIP Advances*, vol. 15, no. 2, p. 025305, Feb. 2025.
7. **Kazi T. Mahmood** and M. Arif Hasan, "Experimental demonstration of classical analogous time-dependent superposition of states," *Scientific Reports*, vol. 12, no. 1, p. 22580, Dec. 2022.
8. Abrar N. E. Faiaz, Donte J. McMullen, **Kazi T. Mahmood**, and M. Arif Hasan, "Page-curve-like entropy dynamics in a classical elastic bit lattice," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 36, no. 4, p. 043127, Apr. 2026.
9. Abrar N. E. Faiaz, Donte J. McMullen, **Kazi T. Mahmood**, M. Arif Hasan, Pierre A. Deymier, Keith Runge, and Joshua A. Levine, "Experimental Geometric State Control on the Bloch Sphere via Linear Mechanical Resonator," Accepted: *Journal of Vibration and Acoustics*, 2026.
10. Abrar N. E. Faiaz, Akinsanmi S. Ige, **Kazi T. Mahmood**, Jake Balla, M. Arif Hasan, Pierre A. Deymier, Keith Runge, and Joshua A. Levine, "Modeling Phi-Bits, Qubit Analogues, in Acoustic Waveguides," Under review: *Applied Acoustics*.
11. Abrar N. E. Faiaz, Donte J. McMullen, **Kazi T. Mahmood**, M. Arif Hasan, and Pierre A. Deymier, "Phase-Pattern Reversals and Topological Analogies in a Symmetry-Broken Coupled Resonator Network," Under review: *Journal of Vibration Engineering and Technology*.

CONFERENCE PRESENTATIONS

- "Classical analogous quantum coherent superposition through the Hertz-type nonlinearity of elastic granules," 184th *Meeting of the Acoustical Society of America*, Chicago, Illinois, USA, 2023 (**Technical Talk**).
- "Decoherence-Resistant Elastic Bits: A Novel Approach to Quantum Analogue Computing Using Coherent Superpositions of States," *International Conference on Smart Mobility and Vehicle Electrification*, Lawrence Technological University, Detroit, Michigan, USA, 2023 (**Technical Talk, Hybrid**).
- Graduate Research Symposium, Wayne State University, USA, 2024 (**Poster Presentation**).
- "Observation of Berry's Phase in Elastic Bits: Quantum Analogues in Classical Granular Systems," *APS March Meeting*, Minneapolis, USA, 2024 (**Technical Talk, Online**).
- "Exploration of the Quantum Coherent States Through Hertz-Type Classical Nonlinearity," *APS Eastern Great Lakes Spring Meeting*, Kettering University, Flint, Michigan, USA, 2024 (**Technical Talk**).
- "Quantum Concepts in Classical Realms: Berry Phases and Elastic Bits in Granular Systems," *APS Eastern Great Lakes Spring Meeting*, Kettering University, Flint, Michigan, USA, 2024 (**Poster Presentation**).
- "Elastic Bits and Berry Phases: Investigating Topological Phenomena in a Classical Granular Network," 186th *Meeting of the Acoustical Society of America/Acoustics Week in Canada*, Ottawa, Ontario, Canada, 2024 (**Technical Talk**).
- "Realization of Berry Phase in Classical Nonlinear Granular Network Using Quantum Analogue Theory," *APS Eastern Great Lakes Fall Meeting*, Marietta College, Marietta, Ohio, USA, 2024 (**Technical Talk**).
- "Bridging Classical Nonlinearity and Quantum Superposition," *International Mechanical Engineering Congress & Exposition*, Portland, Oregon, USA, 2024 (**Technical Talk**).
- "Quantum Realm in Classical Mechanics: A New Basis of Topological Computing," *International Mechanical Engineering Congress & Exposition*, Portland, Oregon, USA, 2024 (**Poster Presentation**).
- "The Notion of Berry Phase in Classical Analogues System," 187th *Meeting of the Acoustical Society of America*, 2024 (**Technical Talk, Online**).
- "Classical Granular System Topology: Berry Phase Formation in Elastic Bit," *International Electronic Conference on Applied Sciences*, 2024.
- NewFoS Research Symposium, University of Arizona, Tucson, Arizona, USA, January 2025 (**Poster Presentation**).
- "Elastic Bits in Granular Metamaterials: A Quantum-Inspired Framework for Robust Topological Computing," 188th *Meeting of the Acoustical Society of America*, New Orleans, Louisiana, USA, 2025 (**Technical Talk**).
- "Modeling Phi-bits in Acoustic Metamaterials for Quantum-Inspired Computation," 188th *Meeting of the Acoustical Society of America*, New Orleans, Louisiana, USA, 2025 (**Technical Talk**).
- "Geometric Phase and Topological Control of Elastic Lattices on the Bloch Sphere," *APS Global Physics Summit*, Denver, Colorado, USA, 2026 (**Technical Talk**).
- "Hilbert space framework for geometric phases and topological dynamics in mechanical lattices," 190th *Meeting of the Acoustical Society of America*, Philadelphia, Pennsylvania, USA, 2026 (**Technical Talk**).
- "Exploring phi-bits in coupled acoustic waveguides: A classical platform for quantum-inspired informa-

tion processing," 190th *Meeting of the Acoustical Society of America*, Philadelphia, Pennsylvania, USA, 2026 (**Technical Talk**).
- "State-Space Geometry and Geometric Phase in Static and Modulated Elastic Lattices," *APS Eastern Great Lakes Section (EGLS) & MIAAPT Joint Meeting*, Wayne State University, Detroit, Michigan, USA, 2026 (**Technical Talk**).

PROFESSIONAL EXPERIENCE

Assistant Teaching Professor **Bowling Green, Ohio, USA**
School of Engineering
Bowling Green State University *Aug 2026 – Present*

- Independently design and deliver four undergraduate engineering courses as the instructor of record: **MME 1100: Basic Computer-Aided Design**; **ENGT 2100: Solid Modeling**; **MME 2400: Statics**; and **ENGT 4250: Manufacturing Design and Operations**.
- Develop course syllabi, lecture materials, laboratory and hands-on activities, assignments, and projects aligned with course learning outcomes and engineering applications.
- Integrate engineering analysis and industry-standard CAD tools into instruction to build students' skills in technical drawing, three-dimensional modeling, mechanics, and design for manufacturing and assembly.
- Design and administer quizzes, examinations, and project-based assessments; evaluate student performance and provide timely, individualized feedback.
- Manage course delivery, grading, student communications, office hours, and academic support while maintaining an inclusive, engaging, and academically rigorous learning environment.

Postdoctoral Researcher **Detroit, Michigan, USA**
Acoustic Pseudospin and Metamaterials Laboratory
Wayne State University *Feb 2026 – Aug 2026*

- **Project 1:** Developed reduced-order models of elastic-bit transfer to optical/RF systems, tracking phase and state trajectories to assess distortion from delay, damping, noise, and interface mismatch.
- **Project 2:** Proposed nonlinear band-edge waveguides for Li-ion battery thermometry, using harmonic and geometric-phase diagnostics to separate internal temperature changes from measurement drift.
- **Project 3:** Developed elastic-bit co-processor concepts for low-power, always-on sensing, using reconfigurable mechanical front-ends for filtering, spectral shaping, and feature extraction before digitization.

Graduate Research Assistant **Detroit, Michigan, USA**
Acoustic Pseudospin and Metamaterials Laboratory
Wayne State University *May 2024 – Dec 2025*

- **Project 1:** Developed the **Elastic Bit**, a classical analog of a quantum bit, in granular-bead elastic media using linearized and nonlinear models. Constructed analytical and experimental nonlinear granular-network models with **asymptotic perturbation techniques**, demonstrating time-dependent superposition responses for potential data transmission (NSF Grant 2204382).
- **Project 2:** Developed a theoretical framework for manipulating elastic-bit superposition states and generating **Berry phase** in linearized and nonlinear elastic networks, demonstrating quantum-classical correlations in time-dependent and time-independent systems (NSF Grant 2242925).
- **Project 3:** Developed an algorithm to evaluate the scalability of the **Logical Elastic Bit** in a mass-conical spring model. Modeled nonlinear harmonic response to quantify time-dependent elastic-bit behavior.
- **Project 4:** Investigated symmetry breaking in granular networks using a **second-order molecular dynamics** algorithm, revealing internal parameter variations that drive topological transitions across the **Brillouin zone**. Studied multi-mass unit cells to explore the corresponding **Hilbert space**.

Research Collaborator **Tucson, Arizona, USA**
New Frontiers of Sound (NewFoS)
The University of Arizona *Sep 2023 – Present*
Project: *Topological Acoustic (TA) Quantum Analogies for Quantum Information Science (QIS)*

- **Project Lead:** Investigated multi-qubit analogues in a classical nonlinear mass-conical spring oscillator. Demonstrated scalable harmonic structures for superposition states and quantum-analogous gate operations, including Hadamard and phase gates (NSF Grant Number 2204400).

- **Co-Investigator:** Developed the **Phi-Bit** (Phase Bit) by studying three coupled finite-length acoustic waveguides. Modeled continuous phase behavior across varying waveguide conditions (NSF Grant Number 2204400).
- **Co-Investigator:** Contributed to the analysis of phase-pattern reversals in a symmetry-broken three-mass resonator network. Studied how reversing the stiffness ordering of coupled springs changes the internal phase/sign structure of the lower nonzero vibrational branch, providing a finite-system analogy to topological band reorganization and nonlinear-mode-like behavior.

Research Experience Mentor **NSF Project**
New Frontiers of Sound (NewFoS) *Jan 2023 – Present*

- Mentored and supervised undergraduate researchers through the NSF New Frontiers of Sound (NewFoS) Research Experience and Mentoring (REM) program, guiding them in hypothesis formulation, experimental planning, data acquisition, and interpretation of results in nonlinear and topological acoustics.
- Trained students in the design and analysis of elastic waveguide systems, including mode identification, dispersion characterization, and phase–amplitude mapping, to investigate classical analogues of quantum information states.
- Facilitated literature review sessions and data presentation workshops to strengthen students' ability to connect experimental observations with theoretical models and communicate findings effectively.
- Oversaw independent research projects culminating in poster presentations at institutional and national conferences, fostering early-career engagement in interdisciplinary acoustics and quantum-inspired mechanics.

Graduate Teaching Assistant **Detroit, Michigan, USA**
Department of Mechanical Engineering
Wayne State University *Jan 2022 – May 2024*

- Served as **Course Instructor** for **ME 3400: Dynamics I** and **ME 5000: Engineering Analysis**, delivering lectures, designing problem sets, and evaluating student performance for both undergraduate and graduate cohorts.
- Led laboratory sessions for **ME 7650: Topological Acoustics**, guiding students through experimental procedures, data analysis, and the interpretation of topological acoustic phenomena.
- Provided instructional support for multiple laboratory and recitation sections, including grading, feedback, and individualized student mentoring.
- Assisted in exam administration and proctoring, ensuring academic integrity and smooth execution of assessments.

Engineer Trainee (Intern) **Chattogram, Bangladesh**
BSRM Steel Limited *Jan 2020 – Feb 2020*

- Gained hands-on experience in industrial steel-rod manufacturing, including billet reheating, multi-pass hot rolling, controlled water quenching, and atmospheric cooling for TMT reinforcement bars meeting ASTM and BSTI standards.
- Assisted in the **mechanical maintenance** of re-rolling mill equipment, including rolling stands, drive shafts, guide assemblies, and cooling-bed conveyors. Performed routine inspections, alignment, lubrication scheduling, and fault diagnostics to reduce unplanned downtime and support production safety.

Undergraduate Research Assistant **Dhaka, Bangladesh**
Department of Mechanical Engineering
Bangladesh University of Engineering & Technology *Nov 2019 – Feb 2021*

- Studied the strength of metal matrix composites (MMCs) subjected to electrical current through thin composite layers. Analyzed current-induced thermal stress and the effects of MMC mesh geometry.
- Analyzed stress tolerance limits and failure characteristics due to thermal and mechanical stresses in the material using a **tensile** and **compression** testing machine.
- Developed a **heat pipe** exchanger model to increase heat transfer efficiency in a compact thermal management system for a high-performance induction motor.

TECHNICAL SKILLS

- **Fabrication & Experimental:** Granular bead networks, mass-conical spring models, coupled linear waveguides, heat pipe exchanger, vibration testing design, topological mass-sensing prototype design.
- **Programming & Simulation:** C/C++, Python, MATLAB, LabVIEW, COMSOL, ANSYS FEA.
- **Mathematical Modeling:** Finite difference and finite element formulation, multi-order asymptotic perturbation modeling, Fourier harmonics, nonlinear dynamics modeling, structural dynamics.
- **CAD & Design:** SolidWorks, AutoCAD, CATIA, Creo, GD&T.
- **Instrumentation & Hardware:** Laser Doppler Vibrometer (Polytec), Oscilloscope (Tektronix), Function Generator (B&K Precision), Power Amplifier (Piezodrive), Vibration Shakers/Exciters, Transducers, Synchronized Sensor Integration.
- **Standards & Methods:** Experimental data acquisition, noise isolation, preventive maintenance, diagnostics, hardware characterization, root-cause troubleshooting.
- **Productivity & Documentation:** LaTeX, Microsoft Office.

ACADEMIC SERVICES

Reviewer

- Scientific Reports, Nature Portfolio — Jan 2024 – Present
- Journal of Vibration & Acoustics, ASME — Jan 2025 – Present
- IDETC-CIE, ASME — 2026
- Discover Quantum Science, Springer Nature — Apr 2026 – Present

AWARDS

- Graduate Student Professional Travel Grant, Wayne State University, 2024.

VOLUNTEERING ACTIVITIES

- Member, American Society of Mechanical Engineers (ASME) — Jun 2024 – Dec 2025
- Member, American Physical Society (APS) — Jul 2023 – Dec 2025
- Member, Acoustical Society of America (ASA) — Jan 2023 – Present
- Executive Member, BUET Robotics Society — Feb 2016 – Jul 2020

REFERENCES

M. Arif Hasan, Ph.D.
Assistant Professor
Department of Mechanical Engineering
Wayne State University
NSF CMMI Panel Fellow
Editorial Board Member at Scientific Reports (Nature)
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Pierre A. Deymier, Ph.D.
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Professor, Department of Materials Science and Engineering
CNRS Fellow Ambassador
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Keith Runge, Ph.D.
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